Jacob Mattie

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Summary

Full-Stack Engineer with a mathematics background. Specialized in data pipelines, databases, and automation.

Python, C++, SQL, Bash | **Linux, Windows** | **Git, AWS, Django**

Work Experience

Investment Analyst / Full Stack Engineer — Vancouver, BC
Aligned Capital Partners — 11/2024 - Ongoing

- Built ETL **data pipelines** merging over 15 sources (files, APIs, web scraping) into a PostgreSQL schema, enabling the use of **machine learning** techniques for investment decisions.
- Configured a **Linux-based server stack** (Fedora, Gunicorn, Nginx) to host a **Django** application built from scratch, making data analytics and automation tools accessible to administrative staff.
- Automated recurring tasks in Python, eliminating over 20 hours of manual work per month and reducing the likelihood of human error.

Test Technologist — Burnaby, BC
IBC Technologies — 6/2022 – 9/2023

- Automated testing with IoT microcontrollers and PLC devices, creating **cloud-connected data pipelines** to send controls and receive data from validation units.
- **Automated data analysis** of high-volume datasets using MATLAB, reducing processing time from 20 minutes to under 3 minutes per iteration.

IT Technician (Internship) — Vancouver, BC
BC Cancer Research Centre — 09/2018 – 01/2019

- Diagnosed and resolved technical issues across Windows, Mac, and Linux environments in a large-scale research facility, showing adaptability and **rapid problem-solving** skills.
- Configured and maintained network infrastructure and digital policy to ensure **cybersecurity** compliance and reliable data transmission across departments.

Education

Simon Fraser University — 01/2019 - 05/2022
B.A.Sc in Mathematics

Volunteering & Leadership

Triumf - ASPIRE Astrochemistry Lab — Vancouver, British Columbia
Automation Technologist — 05/2025 - Ongoing

- Contributed to the publication of 2 papers:
  - Astrochemistry; In Preparation
  - Control Systems; In Preparation
- Developed a Bash-based installer for EPICS, an industry-standard **lab control framework** for physics laboratories, mitigating a significant entry labour barrier for new students and staff.
- Integrated a Python motor controller into the laboratory control system, **eliminating effective experimental variance** and reducing need for active engagement with the experiment over its runtime.